

Agentic Laboratories

The vista with two objects, and the vocabulary the collector needs before it collects anything

The page for this vista is a title, a heading that says Stewing, and one link. That is an honest account of how far the thinking has gone in prose. It is not an honest account of how much of the work exists, because the instrument this vista runs on is already designed, already named, and already has a mock-up of its own in the tools folder. What the site does not carry is the sentence that says why the instrument is the vista rather than a tool sitting inside one.

This file is that sentence and the vocabulary under it.

Contents

1	The vista	2
1.1	What the title names	2
1.2	The question the openings table gives this vista	2
1.3	Why the interaction has to be the record	3
1.4	The other half of the same record	4
1.5	Why five years and not one	4
1.6	The one link on the page	5
2	Active, the instrument	5
2.1	The name and the thing	5
2.2	The four pieces	6
2.3	Consent	8
3	Active, the five dimensions	9
4	Active, the evidence the vista rests on	10
5	Working	12
6	Where this vista meets the artifacts folder	13
7	What this vista leaves open	14
8	Sources	14

1 The vista

1.1 What the title names

Future of Work in Agentic Laboratories: Open Science, Health Scientists.

Two objects follow the colon and they are not one object. Open science is a property of what comes out of the laboratory. The health of the scientists is a property of the people it came out of. Almost all of the current writing about agents in research measures the first and assumes the second arrives with it.

The site says the second one twice. Wellbeing is one of the five dimensions every interaction record is annotated along, and the study question is put in those terms:

“How agents are used in research work. What that use does to research productivity and to research quality. What it does over time to the growth and health of a research community.”

website, Agentic Lab, the study it is for

Three questions, and the third one is the one nobody has an instrument for. A laboratory can raise its output and lose the people who produced it, and the publication record will look identical either way for about two years.

The word laboratory is being used in its social sense here and not its equipment sense. The unit is a research group. The subjects are named:

“Collection from graduate students and postdoctoral researchers in Computer Science first, then more widely across the College of Engineering, with graduate students as the major driving force of the study as well as its subjects.”

website, Agentic Lab, the study it is for

Subjects and drivers at once. That is a design commitment and not a convenience, and it constrains the consent model rather than being constrained by it. A graduate student who is running the study and being studied by it cannot be consented the way an anonymous participant is consented.

1.2 The question the openings table gives this vista

One row on the site assigns a research question to this vista, and it is not the question the artifact pages answer.

“How can we democratize the benefits of agentic solutions? Do people know how to use Copilot or Claude? A investigation of open source MCP eco-system from a Human-AI teaming perspective”

website, Work With Me, the openings table

Read the second sentence literally. It does not ask whether people have access. It asks whether people who have access know how to use what they have. Those are different questions and only the second one is hard. Access is a licensing problem and a procurement problem, and a university solves it with a purchase order. Knowing how to use the thing is a skill-formation problem, and the answer to it varies inside one lab, inside one week, between two sessions of the same person.

That is why this vista needs a collector and not a survey. The quantity it is after is not per person.

It is per session.

The third sentence names the substrate the tools actually arrive on. The open source MCP ecosystem is where a research group's agent capability comes from in practice, one server at a time, installed by whoever needed it that afternoon. The first large study of that ecosystem looked at 1,899 open source servers and found general vulnerabilities in 7.2 percent of them and MCP-specific ones in 5.5 percent, with two thirds carrying code smells. Democratizing access to that is also democratizing exposure to it, and the human-AI teaming perspective the question asks for is the part that decides which of the two a given lab gets.

1.3 Why the interaction has to be the record

The argument is on the artifact page in four sentences and it is the strongest piece of writing in the folder.

“Research work is moving into agents faster than the evidence about its effects is accumulating, and the evidence that exists is not reassuring.”

website, Agentic Lab, why the interaction has to be the record

“Self-report is therefore not a usable instrument. The interaction itself has to be the record.”

website, Agentic Lab, why the interaction has to be the record

Two results stand behind that, and both are already cited on the neighboring vista page in the same order, doing different work there. On the competitive learning page they explain why a competition stopped measuring ability. Here they explain why a questionnaire cannot be the instrument.

The first is the METR randomized trial. Sixteen experienced open source developers, 246 real issues in repositories they maintain. Before the study they forecast that the tools would cut their completion time by 24 percent. After the study they estimated that the tools had cut it by 20 percent. The measured effect was a 19 percent slowdown. Three numbers about the same quantity, two of them produced by the person who did the work, and the sign is wrong in both.

That trial has a sequel, and the sequel changes how the number can be quoted. METR reran the design in early 2026 with mostly new developers at a third of the pay, reported estimated speedups of minus 18 percent for the returning developers and minus 4 percent for the new ones, and then said that its own estimates are an unreliable signal. The reason is selection. Some developers declined to take part because they did not want to work without their tools, and between 30 and 50 percent of those who did take part held back the tasks they expected the tools to help most on. The productivity figure is fragile and I would not build a five-year line on it.

The self-knowledge figure is not fragile in the same way, and that is the one this vista uses. Selection changes who was measured. It does not touch the finding that the people who were measured predicted one sign and got the other, about their own work, in their own repositories, twice.

The second is the skill formation experiment. Fifty-two software engineers learning an unfamiliar asynchronous library, randomized on assistance. The assisted group finished in about the same time and scored 50 percent against 67 percent on the comprehension test afterwards, with the largest drop in debugging. Speed unchanged, understanding down.

A third result in that same study is the one that settles the design. The authors separate six interaction patterns and find that three of them involve cognitive engagement and preserve the

learning outcome even under assistance. Full delegation is the pattern that costs. So the effect of an agent on a researcher is not a property of the researcher and not a property of the tool. It is a property of the interaction, session by session, and it changes within a person across a day.

There is no self-report instrument that recovers that. A person cannot report a pattern they were not aware of adopting, and the METR result says that even the gross direction of the effect is not available to introspection. The interaction is the only place the quantity exists.

That is the entire case for the substrate, and it is a case about measurement rather than about agents.

1.4 The other half of the same record

The evidence problem has a second face that the artifact page does not state, and it is the one that connects this vista to open science rather than to wellbeing.

A survey of autonomous research agents published this year organizes its whole review around one question, which is not whether agents can do science but how a reviewer would know. It looked at 24 runnable systems. 83 percent released code. 38 percent released the seeds or the execution traces that would let somebody reproduce a run. 38 percent reported any method at all for checking that an output was novel. The survey calls the space between what these systems produce and what a reviewer can check the verification gap.

The missing 45 percent is execution traces. That is the same object as the interaction record, arriving from the opposite direction. One vista wants the record because it says something about the people. Open science wants the record because without it a result produced with an agent cannot be audited by anybody who was not in the room.

One substrate, two obligations, and they do not compete. This is why the vista title puts open science and healthy scientists on the same line.

1.5 Why five years and not one

Three reasons, in the order they matter.

The claim is longitudinal by construction. What agent use does to a research community over time is not answerable from a cross-section, and a cohort of graduate students takes four to six years to pass through. Any shorter study measures the first weeks of a habit.

The instrument has to survive its own revisions. A protocol that gets improved mid-study destroys the comparison it was built for unless the version is stored with every record, which is the design commitment the artifact page leads with. A five-year study is a study that expects to revise its protocol at least three times and plans for the revisions to be reconstructible.

And the thing being measured is moving. The competitive learning vista documents a structural break inside one year on a platform with six years of population-scale data. If the ability distribution in the field can move that far that fast, a one-year measurement of research practice is a measurement of one point on a slope whose direction is what the question was about.

1.6 The one link on the page

The only content under Stewing is a link to Methodology of Algorithm Engineering, by Mendling, Leopold, Meyerhenke and Depaire. It is worth saying why it is there, because the connection is not obvious from the title.

That paper is not about agents. It is a methodology paper that a field wrote about its own practice, and it gives an ontology of four elements that algorithm research is made of: algorithmic problems, algorithmic tasks, algorithm designs and algorithm implementations. It places the knowledge produced in Popper's three worlds, and it distinguishes prescriptive from descriptive claims, so that a contribution can be asked what kind of claim it is making before it is asked whether the claim is true.

The relevance to a collector is direct. An interaction record has to attach to something. If a session is annotated only as work, then every measure computed from it is a measure of undifferentiated effort, and the finding that comes out is that people who spend more hours produce more output. The four-element ontology gives the units an annotation can point at, and it separates the interaction where somebody chose a problem from the interaction where somebody made an implementation faster. Those two look identical in a timestamped log and they are not the same event in the science.

It is also the genre this vista belongs to. A field examining its own method, using vocabulary borrowed from philosophy of science, because the internal vocabulary was not sufficient to state the question. That is the shape of the work here, with the difference that the object under examination is a laboratory whose members now work alongside agents.

2 Active, the instrument

2.1 The name and the thing

agentic laboratory

Definition: A research group whose ordinary work has moved into agents, taken as a unit of study. The object is the group and its practice, not the discovery it produces.

Distinguishes it from: A self-driving laboratory, which is an instrumented physical facility closing a loop from hypothesis to robotic experiment to result with the human out of the cycle. The autonomy literature grades those on a scale from scripted automation up to closed-loop discovery. An agentic laboratory in this sense has no such scale, because the human is not being removed from the loop and is the thing being measured.

Operational test: Who is the subject of the study? If the answer is the discovery pipeline, it is a self-driving laboratory question. If the answer is the people and their practice, it is this one.

In this project: The subjects are graduate students and postdoctoral researchers in Computer Science, then the College of Engineering. Every dimension in the annotation scheme except science quality is a property of humans.

Agentic Lab

"A consent-aware substrate for collecting and analysing researcher-agent interaction, across teamwork, cognition, science quality, wellbeing, and organizational dynamics."

website, Agentic Lab, one-line description

Definition: The collection-and-analysis layer for this vista. Four pieces: a typed event schema, multi-provider ingestion adapters, a provenance graph with team, task and cognition annotations, and weekly diagnostics for data quality and drift. A consent gate sits under all four.

Distinguishes it from: The studies that run on top of it. CHART generates one kind of interaction record and Astrolabe codes another. Agentic Lab is the layer those feed into, so that the consent and provenance guarantees are carried through rather than rebuilt once per study.

Operational test: Would removing this component change what a specific study can conclude, or would it change what every study can conclude? The second answer means the component belongs here and not in the study.

In this project: It is the vista’s instrument. Everything the vista claims about a five-year effect is a claim about what this thing recorded.

2.2 The four pieces

event schema

“one typed record per interaction, so heterogeneous sources become comparable”

website, Agentic Lab design, the MVP

Definition: The fixed set of typed fields that every researcher-agent interaction is written into, whatever tool it came from, so that two records from two providers can be counted together.

Distinguishes it from: A security event schema, which is what the phrase means everywhere else in my work. The Open Cybersecurity Schema Framework normalizes security telemetry across vendors with categories, event classes and an attribute dictionary, and the Elastic Common Schema does the same job inside one product family. Both normalize machine events for detection. This one normalizes human-agent exchanges for study, and its hard cases are the ones a detection schema never has: which participant originated a turn, and what the person was trying to do.

Operational test: Can a record from provider A and a record from provider B be placed in the same count without a per-provider correction? If not, the schema has not been fixed yet, it has only been described.

In this project: Listed first in both the MVP and the status section, and the status section calls it fixed.

ingestion adapter

“so interaction that happened across different agent platforms lands in one store”

website, Agentic Lab design, the MVP

Definition: A per-provider component that reads whatever a given agent platform exports and writes it into the event schema.

Distinguishes it from: An adapter in the parameter-efficient fine-tuning sense, a small trainable module inserted into a frozen network. Nothing is learned here. The adapter is a translation layer and its correctness is checked by whether the same interaction, captured through two adapters,

produces the same record.

Operational test: If a provider changes its export format next quarter, how many files change? One adapter, or the schema too? If the schema moves, the schema was written against a provider rather than against the interaction.

In this project: Multi-provider from the start, which is a decision about the population rather than about engineering. A lab does not use one tool, and a study that captures one provider measures the subset of the work that happened to land there.

provenance graph

“the interactions linked into a structure you can trace and query”

website, Agentic Lab design, the MVP

Definition: The linked structure over interaction records, annotated with team, task and cognition, so a result can be traced back through the sessions that produced it.

Distinguishes it from: Two established senses, and they are not the same thing. In my own field a provenance graph is the causal graph built from system audit logs, processes and files and sockets with edges for the operations between them, and it is what a detector for advanced persistent threats runs over. In open science, provenance is the W3C PROV sense, entities and activities and agents, recording what produced a piece of data so its reliability can be assessed. This one is the second. It is a lineage record for research work, and its agents happen to include language models.

Operational test: Is the graph built from telemetry the participants did not author, or from a record of what they did with intent? An audit-log provenance graph is the first. This one is the second, and that difference is why it needs a consent gate and the security one does not.

In this project: The unit the five dimensions annotate. The annotation lives on the graph rather than on the raw event, which is what lets one interaction be read as a coordination event and a cognitive step at once.

diagnostics

“because a collection substrate that silently degrades is worse than none”

website, Agentic Lab design, the MVP

Definition: The weekly automated checks on data quality and drift over the collected records.

Distinguishes it from: Diagnosis in the CHART sense, where the object being diagnosed is a team’s coordination. Here the object is the collection itself. The subject of the sentence is the instrument, not the participants.

Operational test: If an adapter started dropping every third tool call on Tuesday, when would anyone find out? A collector with no answer to that has a five-year study with an unknown hole in it.

In this project: Weekly, and listed as fixed. The justification is stated in the strongest terms on the page, that a substrate degrading in silence is worse than not collecting at all, which is correct: an absent record is a known gap and a degraded record is a false one.

drift

Definition: Change over time in what the collection is capturing, as distinct from change in what the researchers are doing.

Distinguishes it from: Drift in the Toward Deployment sense, which is distribution shift that matters to a specific deployed policy. That drift is a property of the world moving away from a model. This drift is a property of the instrument moving away from itself. The two need separating because in a five-year study they produce the same signature, a measure that changes, and only one of them is a finding.

Operational test: Would the measured change appear if every participant behaved identically to last year? If yes, it is instrument drift.

In this project: Named alongside data quality in the weekly diagnostics, which is where it has to be caught, because there is no way to recover it after the fact from the records themselves.

2.3 Consent

consent gate

“block responses are saved only under an explicit, versioned protocol, so the exact instrument used for every session is preserved even as later protocol versions appear”

website, Agentic Lab, the shape of it

Definition: The condition on capture. A response is stored only under an explicit consent protocol, and the version of the protocol is stored with the data.

Distinguishes it from: A consent checkbox, which is a one-time event at enrollment and which cannot say what a given record was collected under after the form has been revised. It is also not a control dependency in the CHART sense, where an action waits for authorization from a teammate before it fires. Both are gates and both hold an action until permission exists, but a control dependency governs what an agent may do inside a task, and this governs what the researcher may keep afterwards.

Operational test: Pick any stored record at random. Can the exact instrument and the exact terms under which it was collected be reconstructed from what is stored, without consulting anyone’s memory? If not, there is no gate, only a checkbox.

In this project: The page calls it the design commitment that shapes the rest, and it does shape the rest. It is why the schema carries a protocol version field, and it is what makes the longitudinal claim defensible rather than a moving target.

protocol version

Definition: The identifier of the consent instrument in force when a record was captured, stored on the record.

Distinguishes it from: A protocol in the network sense, which is what the word means in every other file in this folder. Also from a study protocol in the registration sense, which is the plan for the analysis rather than the terms of the capture.

Operational test: Two records six months apart show the same field with different values. Is

that a change in the participant or a change in what the question asked? Without the version, the two are indistinguishable and the longer the study runs, the more of the data has this problem.

In this project: This is the field that buys the five years. Everything else in the design could be rebuilt later from stored raw data. This one cannot, because the information it holds stops existing the moment the protocol is revised.

3 Active, the five dimensions

Every record on the provenance graph is annotated along five axes, and the axes are definitions rather than labels. Each is one clause on the design page.

teamwork

Definition: The coordination structure of the interaction.

Distinguishes it from: Taskwork, the content of what was done. The Human-AI Complementarity cluster already splits these, and this dimension is the same split applied to research work instead of to network defense.

Operational test: Would the record change if the same actions had been taken by one party alone in the same order? If not, nothing on this axis was measured.

In this project: The dimension CHART traces feed. The dependency vocabulary that cluster defines is available here without redefinition.

cognition

Definition: The reasoning and problem-solving steps inside the interaction.

Distinguishes it from: Output quality, which is what the steps produced. Astrolabe exists because these two come apart: a correct answer reached by a route nobody can reconstruct is a bad research record and a good result.

Operational test: Can the annotation be made from the final artifact alone? If yes, it is a quality annotation wearing a cognition label.

In this project: The dimension Astrolabe codes, and the one with an existing scheme, an existing agreement measure, and existing coders.

science quality

Definition: Whether the work meets the standards it claims.

Distinguishes it from: Productivity, counted separately in the same study question. The pairing is deliberate and the vista's whole worry is that the two move in opposite directions under agent use while only the first is visible.

Operational test: Take the claim the work makes and ask what would have to be true for it to hold, then ask whether the record shows that check being done. This is the verification gap question asked at session scale rather than at system scale.

In this project: The one dimension of the five that is a property of the output rather than of the person. It is also the dimension that connects this vista to open science.

wellbeing

Definition: The human's state through the interaction.

Distinguishes it from: Satisfaction with the tool, which is what a product survey measures and which the METR result shows is not a proxy for anything. The developers in that trial were slowed and were pleased.

Operational test: Is the measure available from the interaction, or does it require asking afterwards? Both kinds are admissible, but only the first survives the self-report problem, so a study that has only the second kind on this axis has inherited the weakness the design was built to avoid.

In this project: Half of the vista's title. It is also the axis with the weakest instrumentation, and saying so is not a criticism of the design, because nobody has a good one.

organizational dynamics

Definition: How the interaction sits inside larger structures of authority and process.

Distinguishes it from: Teamwork, which is coordination among the participants in the session. This is the structure outside the session that decided what the session was for. An advisor's deadline is not in the room and is in the record.

Operational test: Could the participants have changed this by agreeing among themselves? If yes, it is teamwork. If no, it is organizational.

In this project: The axis that makes the study about a laboratory rather than about a set of individuals who happen to share a building.

Why all five sit on one record

The design page states the reason, and it is a claim about the world rather than a convenience of storage.

“a coordination breakdown is often a wellbeing signal, and a cognitive shortcut is often an organizational pressure”

website, Agentic Lab design, five dimensions

The consequence for anyone building an analysis on top: a finding on one axis is not evidence for that axis until the others have been looked at on the same records. A drop in cognitive engagement across a lab in March is a skill-formation finding, a workload finding, or a finding about when the qualifying exam is, and the three are told apart by the annotations sitting next to it rather than by argument.

This is also the reason the five cannot be split into five instruments. Five separate collections, each with its own consent terms and its own sampling, cannot be joined on the record afterwards, and the joining is where the content is.

4 Active, the evidence the vista rests on

self-report

Definition: A measure of research work produced by asking the researcher about it afterwards.

Distinguishes it from: An interaction record, which is produced by the work itself. The distinction is usually presented as one of accuracy. The METR result makes it stronger than that: the self-reported and measured values had opposite signs, so the self-report was not a noisy version of the quantity, it was a different quantity.

Operational test: Does the participant have to remember anything to produce the value? If yes, it is self-report, whatever it is called.

In this project: Ruled out as the primary instrument on the artifact page in one sentence. It is not ruled out as a secondary measure, and the gap between what somebody reports and what the record shows is itself one of the more interesting quantities this substrate can produce.

delegation

Definition: The interaction pattern in which a task is handed to an agent without the human engaging with how it was done.

Distinguishes it from: Assisted work in general, which is the category the whole debate is usually conducted in and which the skill formation study shows is too coarse to carry a finding. Six patterns, three of them preserving the learning outcome. The variable is not whether the agent was used.

Operational test: Did the human ask the agent anything that was not a request for output? Follow-up questions, requests for explanation, conceptual questions asked while working independently. The presence of those is the difference between the patterns that cost understanding and the ones that do not.

In this project: This is the quantity the collector exists to measure, and it is measurable from the interaction and from nowhere else. It also converts the vista's second question, do people know how to use these tools, into something with an answer: the distribution over patterns, per person, over time.

verification gap

Definition: The distance between what an agent-assisted result claims and what a reader can check. In the survey that names it, code release is now standard practice while the artifacts that let somebody re-run a result are not: 83 percent released code, 38 percent released seeds or execution traces.

Distinguishes it from: Reproducibility in the usual sense, which is a property of a released artifact. This is a property of a review, and the survey's organizing question is not whether the agent did science but how a reviewer would know.

Operational test: Hand the paper and everything released with it to a competent stranger. Can they tell whether the result holds, without asking the authors? If the answer depends on a session nobody recorded, that session was the missing artifact.

In this project: The open science half of the vista title. The record this substrate collects is the same object the survey found missing, which means the consent infrastructure built for the wellbeing question is also the infrastructure for the auditing question.

autonomy level

Definition: A graded description of how much of a research process an agent carries without a

human decision in the loop. The published scales run from assistive tool use, through autonomy over one bounded stage, through a chain of stages from idea to write-up, up to a closed loop where the agent revises its hypotheses on its own results.

Distinguishes it from: Capability. A level says what the system is permitted and structured to do unsupervised, not what it does well. The survey that grades systems this way found nine at the closed-loop level and exactly one whose loop closed on an external check rather than on its own metric, and that one predates language model agents.

Operational test: Remove the human for one full cycle. What still happens? The answer is the level, and it is not affected by whether the output was any good.

In this project: Peripheral to the measurement and central to the framing. This vista is not about laboratories at high autonomy levels. It is about laboratories at the low ones, which is where all the people are, and which the autonomy literature treats as the starting point rather than as the subject.

democratization

Definition: In this vista's question, the distribution of the benefit of agentic tools rather than the distribution of the tools.

Distinguishes it from: Access. Access is settled by a site licence. Distribution of benefit is settled by which interaction pattern a person falls into, which is a skill, which is unevenly held, and which nobody currently teaches.

Operational test: Two labs with identical tool budgets. Does the measure separate them? A measure that cannot is measuring access.

In this project: The first sentence of the vista's assigned question, and the sentence that makes it a vista rather than an instrument paper. If the benefit concentrates in the people who were already good at using these systems, the tooling widens the distribution it was bought to narrow.

5 Working

Terms that have to be recognized precisely here, with lookup acceptable.

self-driving laboratory

Definition: An instrumented facility that closes a loop from hypothesis to automated experiment to analysis, with graded autonomy scales published for materials, chemistry and soft matter.

Distinguishes it from: The agentic laboratory of this vista, above. Sharing the word laboratory is the whole of the overlap.

AI scientist

Definition: A system that carries a research process end to end, from question through experiment to write-up.

Distinguishes it from: The agents in this vista, which sit inside a human's work rather than replacing it. Almost all published evaluation effort has gone to the first, which is why the evidence about the second is as thin as the artifact page says.

Model Context Protocol

Definition: The open protocol by which an agent reaches external tools and data, and the ecosystem of servers implementing it. Named in the vista's assigned question as the object of investigation.

Distinguishes it from: The agent itself. Capability arrives through the servers, so a study of what a laboratory can do with agents is partly a study of which servers that laboratory installed.

open science

Definition: Here, the commitment that the record of how a result was produced travels with the result.

Distinguishes it from: Open access, which is about who can read the paper. This vista's sense is about who can check it, which is the verification gap entry above.

inter-coder agreement

Definition: The measured extent to which independent annotators assign the same code to the same material, reported as percent agreement and as a chance corrected coefficient.

Distinguishes it from: Correctness. Astrolabe's position is the one to keep: where coders diverge is treated as a finding rather than as a defect. Any annotation this substrate carries on cognition inherits that.

6 Where this vista meets the artifacts folder

Four tools sit in that folder and three of them belong to this vista in different ways.

"This is the layer beneath the specific studies. CHART generates one kind of interaction trace; Astrolabe codes another; Agentic Lab is the general substrate those specific instruments would feed into, with the consent and provenance guarantees carried through rather than reinvented per study."

website, Agentic Lab design, where it connects

That sentence is the architecture and it is also a claim about my own vocabulary that needs checking, because interaction trace is doing more work in it than the word can carry.

Trace means three things across my own files

CHART uses trace for the linked Input-Process-Outcome record of a human-AI team session. The unit is a session and the participants are a person and an agent working on one mission. CyberAgentFlow uses execution trace for the archived record of a single agent's tool use, prompts and model responses and command outputs and timing, which an analysis model then inspects for inefficiency. The unit is a run and there is no second participant. Agentic Lab uses interaction trace for both of these as inputs, in the sentence quoted above. The third use is the superset and it is fine as long as the schema does not inherit the ambiguity. It cannot, because a CHART record has a coordination structure with two parties in it and a CyberAgentFlow record does not, and any count that pools them is counting two different things. The typed schema is exactly where this gets settled: one record type per unit, with the

party structure explicit, and interaction trace kept as the folder name rather than as a field.

What is built and what is scoped

The two pages of the artifact do not agree on where the tool is.

The README says the build is under way, and lists four things as fixed: the event schema, the multi-provider ingestion adapters, the provenance graph with team, task and cognition annotations, and the weekly data-quality-and-drift diagnostics.

The design page opens by saying this is the intended design of a tool still on the waitlist, and that what follows is the MVP scope rather than a shipped system.

Both can be true of the same object at different dates and the pages carry the same month. Fixed and shipped are different claims, and the README's list reads as design decisions that have stopped moving rather than as running code. Say the narrow version out loud anywhere this comes up: the schema, the adapters, the graph and the diagnostics are specified and settled, and the system is not deployed. That sentence is defensible under either page.

Daedalus is the fourth tool in the folder and it does not belong to this vista at all. It is network emulation infrastructure for training and evaluating defense agents, and it is listed here only so that the boundary of the vista is drawn where it actually falls.

7 What this vista leaves open

The collector is specified. The study around it is not, and three things have to be settled before the first record is worth keeping.

Which unit of research work an interaction attaches to. The algorithm engineering ontology gives four candidate units for one field, and a College of Engineering is not one field. Without a unit, the annotation reduces to effort.

What a wellbeing measure taken from an interaction actually is. Four of the five dimensions have either an existing scheme in a neighboring project or a straightforward operationalization. This one has neither, and importing a questionnaire onto this axis reintroduces the instrument the design rejected.

How a graduate student consents to being studied by a project they are running. The site names them as both subjects and the driving force. That arrangement is the right one for the science and it is the hardest one for the consent gate, and the answer is not a longer form.

The last one is the one I would settle first, because it is the only one where getting it wrong is not recoverable later.

8 Sources

- Every quoted sentence marked *website* is from the vista page under 5-year-vista/, from content/artifacts/agentive-lab/README.md and design.md, or from the openings table on work-with-me.md. The mission and layer list on content/tools/agentive-laboratory-collector-mockup.md agrees with both artifact pages and adds role-based visibility and export and deletion controls to the governance side.

- The competitive learning vista under the same `5-year-vista/` folder supplies the structural break and cites the same two studies in the same order.
- Mendling, J., Leopold, H., Meyerhenke, H., and Depaire, B. Methodology of Algorithm Engineering. arXiv 2310.18979. The one link on the vista page.
- Becker, J., Rush, N., Barnes, E., and Rein, D. Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity. METR, arXiv 2507.09089. Sixteen developers, 246 issues, forecast 24 percent, estimate 20 percent, measured 19 percent slowdown. The design update of 24 February 2026 on the same site supplies the selection problems and the minus 18 and minus 4 percent estimates from the rerun.
- Shen, J. H., and Tamkin, A. How AI Impacts Skill Formation. arXiv 2601.20245. Fifty-two engineers, six interaction patterns, three preserving the learning outcome.
- Autonomous Research Agents: A Survey of AI Scientists and the Verification Gap. arXiv 2608.05179. Twenty-four runnable systems, the release percentages, and the autonomy grading used in the autonomy level entry.
- From AI for Science to Agentic Science: A Survey on Autonomous Scientific Discovery. arXiv 2508.14111, for the task-level and goal-level autonomy distinction.
- Model Context Protocol at First Glance: Studying the Security and Maintainability of MCP Servers. arXiv 2506.13538, later in ACM TOSEM. 1,899 open source servers and the vulnerability rates.
- The W3C PROV data model, entities and activities and agents, and the Open Cybersecurity Schema Framework, for the two established senses in the provenance graph and event schema entries.
- Cluster terms used without redefinition here, taskwork, coordination structure, control dependency, are defined in `hac_shared.tex`. The Toward Deployment sense of drift is in `td_shared.tex`.